

WIND FARM NETWORK STUDY

A second path for SCADA, crews and cameras.

We modelled Jeffreys Bay's terrain from satellite elevation data and checked every radio path against it. two ridge high-sites reach 1 of the 1 mapped substations and give line of sight to 59 of 60 turbines, fed from a carrier uplink on mapped mast, operator not tagged.

Prepared for

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Date

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--- Carrier uplink
 — Private backbone
 — Site link
 ■ Mapped turbine
■ Clear line of sight for a 3 m field terminal

Sentinel-2 imagery, 3 May 2026. The dotted outline is the study area around the turbines. Site positions are desktop candidates, to be confirmed on a site visit.

59 of 60

turbines in line of sight
of the standby
backbone

2 + 0

ridge high-sites plus
short relays

76%

of the 85 km² study
area in line of sight

2.0 km

clear carrier uplink into
the site

Where things stand today

Jeffreys Bay Wind Farm runs 60 turbines across eight farms between Jeffreys Bay and Humansdorp. Everything that tells the control room what the site is doing rides on the fibre. We propose an owned standby wireless path alongside it.

WHAT THE PUBLIC RECORD SHOWS

- 60 Siemens 2.3 MW turbines on about 3,700 ha across eight farms, commissioned in 2014
- SCADA, crew communications and cameras depend on the site's fibre
- The public record describes no standby path for SCADA if the fibre is cut or stolen

Carrier in, private network out

The fibre stays the primary path. An owned wireless path runs alongside it and takes over automatically when the fibre is cut or stolen.

LAYER 1 · CARRIER

Uplink from mapped mast, operator not tagged

An uncontended business-grade Vodacom service, delivered by CTTX over a 2.0 km link to High Site West. It gives the site an independent route out when the fibre route is down.

LAYER 2 · BACKBONE

Ridge high-sites

High Site East (174 m) and High Site West (383 m) sit 7.9 km apart with a clear link between them. Each site runs on solar and lithium storage with remote monitoring, so the network stays up through load-shedding.

LAYER 3 · DISTRIBUTION

Substation, O&M and turbine strings

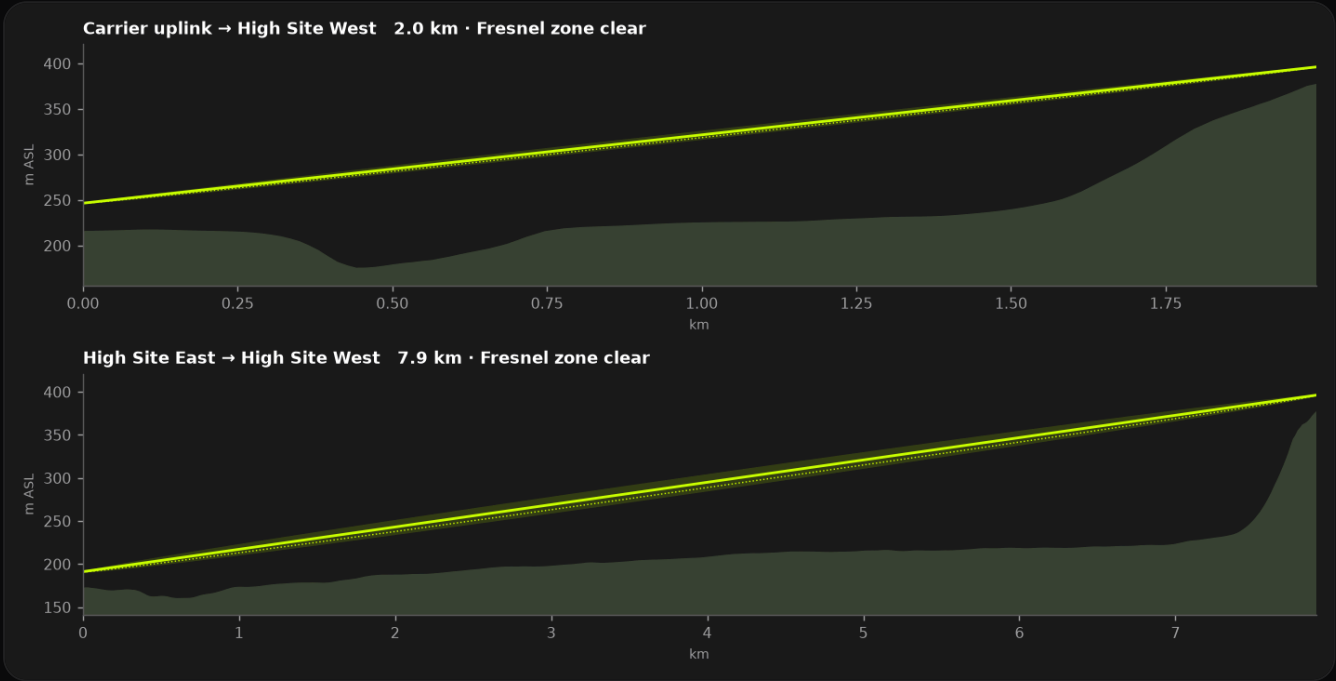
Sector radios reach the substation, the O&M building and the turbine strings, so SCADA, crew radios and perimeter cameras keep reporting on the standby path.

BUILT FOR WHAT COMES NEXT

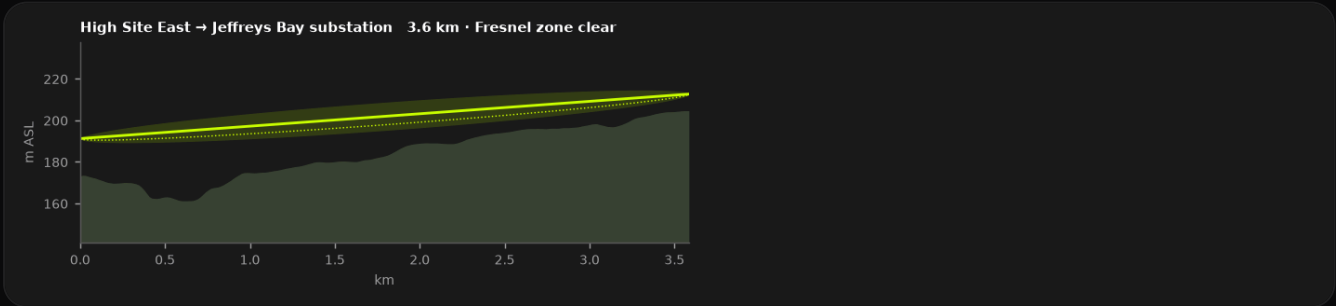
Once the standby backbone covers the site, extra cameras, met-mast data or crew devices are endpoints on a network that is already there, not a new project.

03 Every path checked against the terrain

Each profile shows the ground between two sites, the radio path (lime line) and its Fresnel zone (lime band). We allow for earth curvature and require at least 60% of the zone to be clear at every point. A path that fails is not drawn.



SITE	SERVED FROM	PATH LENGTH	LINE OF SIGHT
Jeffreys Bay substation	High Site East	3.6 km	CLEAR



5.8 GHz · $k = 4/3$ · uplink tower 30 m · high-site masts 18 m · site masts 8 m. Turbine and substation positions come from OpenStreetMap.

04 One standby path for every critical load

A wind farm network carries three loads at once. The standby path is designed to hold all three when the fibre is down.

SCADA CONTINUITY

An automatic second path keeps turbines and the substation visible to the control room when the fibre is cut or stolen.

CREW COMMUNICATION

O&M crews reachable across the site, at the turbines and in the substation, on one monitored network.

SECURITY

Perimeter and substation cameras keep reporting during exactly the incidents that cut the fibre.

05 Where this has been done

CTTX EXPERIENCE

CTTX has delivered this kind of work at Oya Wind Farm and Eskom Medupi.

The same engineering runs an owned backbone on a 30,000 ha reserve:

Sandymount Safaris 30,000+ ha, three lodges

BEFORE

A consumer wireless service at about 3 Mbps down and 8 Mbps up, with the provider on site daily to swap IPs and reinstall equipment.

AFTER

A Vodacom-backed 100 Mbps symmetrical link into the reserve, a CTTX-built main mast and two secondary masts, solar and lithium power, all monitored remotely.

06 What it protects

On a wind farm the case is not a lower bill. It is keeping SCADA visible, crews reachable and cameras reporting when the fibre is cut or stolen. A standby wireless path costs far less than a day of lost visibility.

SIZE IT TO YOUR HISTORY

Tell us how often the fibre has been cut in the last two years and how long SCADA was blind each time, and we will size the standby path for Jeffreys Bay to match.

BUILT TO BE FORWARDED

If this decision sits with someone else, please pass it on. The asset manager will find the availability case, the site manager the crew and O&M case, and security the camera case.

07

What changes with an owned standby path

TODAY

- 60 Siemens 2.3 MW turbines on about 3,700 ha across eight farms, commissioned in 2014
- SCADA, crew communications and cameras depend on the site's fibre
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WITH AN OWNED STANDBY PATH

- SCADA keeps reporting when the fibre is cut
- Crews reachable across the site on one network
- Perimeter cameras stay online during an incident
- An owned asset the site controls, monitored remotely

Avoided loss

Cable and fibre theft can take out SCADA and the cameras in the same incident. A standby path keeps both reporting while the fibre is repaired.

Operational efficiency

Crews keep working across the site when the fibre is down, and faults are seen remotely before a vehicle is sent.

Infrastructure ownership

The standby masts, radios and power belong to the site. They are monitored remotely and independent of the fibre route.

Executive risk reduction

The asset manager gets a documented second path for SCADA and a record of operational diligence for lenders, insurers and the grid operator.

1

30-minute call

How often the fibre has been cut, how long SCADA was blind, and where crews lose contact on site.

2

Confirm the sites

The O&M building, the substation and the fibre route on a marked map. We add them to the model and rerun the link plan.

3

One day on site

We walk the high sites, confirm the uplink with Vodacom and check mounting at the substation.

4

Design and proposal

A final design in Cambium LINKPlanner, with equipment, power, a phased plan and the investment case.

Method. Terrain: Copernicus DEM GLO-30 (30 m). Imagery: ESA Sentinel-2 L2A true colour, 3 May 2026. Radio model: 5.8 GHz, effective earth radius factor 4/3, 60% first-Fresnel-zone clearance at every point. The terrain model does not include trees or buildings. Every path is confirmed on survey before anything is built.

Candidate positions. High Site West -33.97800, 24.80245; High Site East -34.01445, 24.87590; carrier uplink -33.99558, 24.80685 (mapped mast, operator not tagged). The exact carrier site is subject to Vodacom feasibility.

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